TIME SERIES ANALYSIS PROJECT 2024

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Abstract. This report analyzes air temperature data of Højbakkegaard in Denmark using ARMA, Box-Jenkins, and Kalman filter models to model and predict a time series. The methods and functions used are from the course literature[1]. Starting with in part A, with construction of ARMA for temperature. In part B a Box-Jenkins model with net radiation as input in part B and in part C we use the Kalman filter for parameter estimation and prediction refinement for the model we found in part B. Results are validated and compared across models in predicting three different test periods and with a naive predictor for given periods.

"To see a world in a grain of sand,
And a heaven in a wild flower,
Hold infinity in the palm of your hand,
And eternity in an hour."

- William Blake, *Auguries of Innocence*

The raging chief in frantic passion lost,
Blind to himself, and useless to his host,
Unskilled to judge the future by the past,
In blood and slaughter shall repent at last.
"

- Alexander Pope, *The Illiad*

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1. Introduction

The use of accurate modelling and prediction of weather parameters, such as air temperature predicted in this project, to create accurate weather forecast are essential for making informed decisions in areas such in agriculture, energy management, disaster mitigation to name a few. This project focuses on modelling and predicting the local outdoor air temperature in H jbakkegaard, Taastrup, based on data collected there for a number of years. The dataset includes hourly measurements of temperature and other factors such as net radiation, earth heat ux, and re ected shortwave radiation. The analysis aims to apply basic time series techniques learnt in the course to catch meaningful patterns, achieve accurate predictions, and provide insights into temperature dynamics but most importatly hopefully creating an accurate outdoor temperature predictor for H jbakkegaard,

The project is structured into three parts:

Part A: The air temperature data is analyzed independently using the Autoregressive Moving Average (ARMA) model. This step establishes a basic understanding of the temperature series and provides an initial predictive model. Techniques such as Box-Cox transformation, stationarity analysis, and di erentiation are applied to ensure the suitability of the data for ARMA modelling.

Part B: The ARMA model is expanded to incorporate an auxiliary input variables, speci cally net radiation, using the Box-Jenkins (BJ) methodology. By using this approach leverages cross-correlation analysis to identify the relationship between temperature and net radiation with aim to achieve a more accurate and robust model.

Part C: The BJ model is further re ned using the Kalman lter, a recursive estimation method. This step allows dynamic update of parameters and improved predictions over time, especially when there are none stationary periods over time.

Since the dataset spans over several years, the modelling process begins by identifying a relatively stable subset of 8 weeks to serve as the training data. The model's performance is then validated on subsequent weeks and tested on additional datasets, including those from di erent seasons. By using this approach ensures that the model is more robust and generalizes well across varying conditions.

This report presents the methodology and reasoning when nding the model. The methods and functions used are from the course literature[1], which is the main course literature of our study. The are also results graphs and evaluation for each step of the project to improve mind ow. All of the predictors of the models are compared for di erent step size as well a naive predictor.

2. Part A: ARMA Model

2.1. Modeling Using ARMA.

2.1.1. *Data Presentation and Transforming.* We first plot the temperature for all the data given, by inspection of Figure 1 we see that the data is not stationary over time since as expected the temperature between summer time and winter time are different.

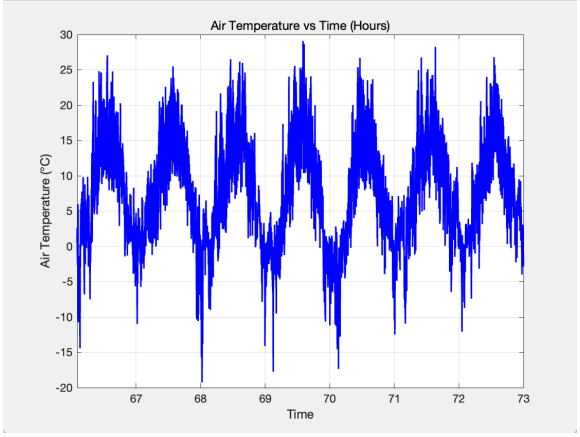

Figure 1. Air temperature vs. hours

Before we proceed to modelling, we first need need to determine whether the whole data set needs to be transformed by applying the Box-Cox function. The Box-Cox transformation Figure 2 shows that it attains maximum at $= 0.95$ which is sufficiently close to 1. We conclude that transformation of data is unnecessary.

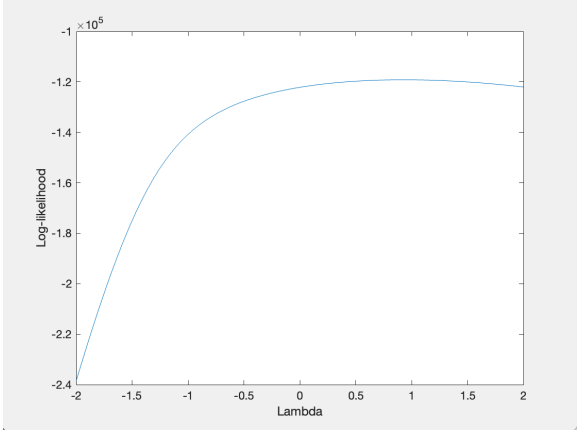

Figure 2. Box-Cox transformation

2.1.2. *Finding Stationary Period.* An 8-week stationary period was selected by examining the rolling standard deviation of the temperature data using a window size